Enhancing Intrusion Detection in CPS and IIoT with Lightweight Explainable AI Models

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2025 IEEE 21st International Conference on Factory Communication Systems (WFCS) | 979-8-3315-3005-1/25/$31.00 ©2025 IEEE | DOI: 10.1109/WFCS63373.2025.11077567

***Abstract*—Integrating cyber-physical systems and the Internet of Things into industrial operations has significantly improved automation, efficiency, and data-driven decision making. However, these advances have also made industrial environments more vulnerable to cybersecurity risks. Our previous work explored lightweight deep learning models for real-time intrusion detection systems on edge devices, yet these models often operate as black boxes, limiting their trustworthiness. This issue is especially critical in the European Union, where the AI Act mandates transparency, accountability, and human oversight for AI solutions to be interpretable. In this paper, we integrate explainable AI solutions into lightweight real-time intrusion detection systems on edge devices to enhance the transparency and interpretability of black-box models. The study demonstrates that integrating SHapley Additive exPlanations significantly enhances the interpretability of intrusion detection systems, providing more transparent insights into model decision-making processes while maintaining accuracy and computational efficiency. This work contributes to the development of more secure and trustworthy industrial ecosystems by improving the effectiveness and reliability of intrusion detection.**

***Index Terms*—Explainable AI, Cyber-Physical Systems, Industrial Internet of Things, Intrusion Detection Systems, Edge Computing, TensorFlow Lite, Machine Learning, IoT Security**

I. Introduction

The adoption of Cyber-Physical Systems (CPS) and the Internet of Things (IoT) in industrial settings has transformed operations by enabling greater automation, improved efficiency, and data-informed decision-making. The Industrial Internet of Things (IIoT) enables real-time monitoring, predictive maintenance, and operational insights by connecting devices, machinery, and systems. However, this connectivity exposes industrial networks to cyber threats, making them vulnerable to attacks that disrupt operations and compromise safety [1]. Traditional Intrusion Detection Systems (IDS) often lack the interpretability needed to comply with standards like the European Union (EU) Artificial Intelligence (AI) Act [2], [3]. Thus, robust and interpretable IDS are critical in IIoT environments [4].

Deep learning models are powerful tools for detecting and preventing cyber intrusions by identifying complex patterns in datasets [5]. However, these models often operate as black boxes, offering little insight into their predictions, which hinders trust and acceptance in security-sensitive applications [6]. While Explainable AI (XAI) techniques improve interpretability, their application to lightweight models tailored for IIoT environments remains limited, especially where computational efficiency and real-time decision-making are crucial.

Our previous work demonstrated the feasibility of deploying lightweight deep learning models, such as convolutional neural networks (CNN), long short-term memory (LSTM) networks, and graph convolutional networks (GCN), on edge devices in IIoT environments [7]. These models addressed the latency and bandwidth challenges of cloud-based processing, but lacked clear decision-making processes [8]. This paper builds on that work by integrating XAI techniques with lightweight models to ensure transparent and justifiable decision-making, essential for trust in automated security operations.

More specifically, this study focuses on integrating SHapley Additive exPlanations (SHAP) [9] with lightweight deep learning models for real-time intrusion detection on edge devices. SHAP provides robust explanations for model predictions by attributing outputs to individual features, offering global and local interpretability. This integration enhances IDS transparency and trustworthiness, supporting system improvements and user confidence in IIoT security. The research questions investigated in this work are,

1) How can explainable AI techniques (i.e. SHAP) be integrated with deep learning models (i.e. CNN, LSTM, and GCN) to improve IDS transparency and interpretability in IIoT environments?
2) What is the impact of incorporating explainable AI on the accuracy, trustworthiness, and efficiency of IDS deployed on edge devices within CPS and IIoT networks?

The remainder of this paper is organized as follows. Section II provides a summary of selected related works. Section III gives an overview of our approach. Section IV describes the method we used. Section V details the implementation and application of XAI to our existing framework. Section VI presents the results of the application of XAI and the measurements obtained. Section VII discusses the findings of our study. Finally, Section VIII concludes the paper by summarizing the research findings and suggesting future work.

II. Related Work

The integration of CPS and IIoT has introduced new security challenges that require robust IDS. CPS and IIoT networks are often characterized by distributed architectures, heterogeneous devices, and real-time operational constraints, making them vulnerable to a wide range of cyber threats [10], [11]. Previous

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studies have addressed these challenges using traditional IDS methods, such as signature-based and anomaly-based detection, which focus on identifying known attack patterns and unusual behaviors [12]. However, these methods often lack the scalability, adaptability, and interpretability needed for modern IIoT environments [6]. Advances in machine learning and deep learning have demonstrated effectiveness in detecting complex attack patterns in CPS and IIoT systems [13]. Lightweight deep learning models, including CNNs, LSTMs, and GCNs, have been proposed to overcome the limitations of centralized IDS by enabling real-time detection in edge devices [7], [14]. Despite their promising results, the black-box nature of these models presents challenges to transparency and trustworthiness, especially in security-critical applications [15], [16]. To address this issue, XAI techniques, such as SHAP [9], have been widely adopted to provide explanations at the feature level of the model predictions. SHAP has been recognized for its ability to integrate interpretability between different machine learning models by attributing importance values to input features [17]. However, recent research highlights certain limitations of SHAP, particularly regarding the stability of its explanations in certain scenarios [18]. Enhancements in SHAP, such as layer-wise neighbor selection, have been proposed to improve the consistency of feature attributions without sacrificing accuracy, making them more suitable for real-time applications in resource-constrained environments [18].

In addition, hybrid IDS frameworks that combine traditional detection methods with deep learning and XAI have been explored to improve detection accuracy and interpretability [19]. These approaches leverage the strengths of different models to create robust, adaptable systems capable of handling the diverse and dynamic nature of IIoT networks. Our previous work demonstrated the feasibility of deploying lightweight deep learning models directly on edge devices within IIoT environments, addressing latency and bandwidth limitations [7]. Although previous work has focused on either lightweight model performance or XAI integration separately, few have explored real-time interpretability on edge-constrained devices. This work bridges that gap by showing how SHAP can be applied practically to CNN, LSTM, and GCN models within an IoT edge-to-cloud architecture.

III. Approach

Fig. 1 illustrates the system architecture, which integrates XAI techniques into our edge computing framework for IIoT environments. The model includes a multilayered approach in which real-time data is processed on edge devices, analyzed for intrusions using lightweight deep learning models, and passed through SHAP for interpretability. The fog layer supports computationally intensive XAI tasks, providing detailed explanations that are then relayed back to system administrators. This architecture ensures that both the detection and the explanation processes are streamlined and efficient, even under the constraints of edge devices. We use edge devices for real-time data collection and processing, which can be attached directly to industrial machines, vehicles, and equipment. This

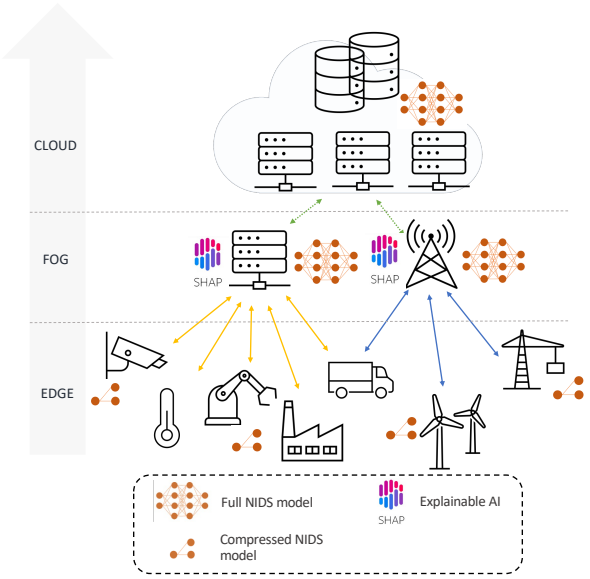


Fig. 1: System architecture

is especially useful when having a stable network connection is difficult. For instance, in a factory scenario, each machine could host an edge device that continuously monitors and collects data regarding the machine's operation, its performance, and the surrounding environment. Access to data right at the source allows for quick data capture and processing, lowering the delay and reducing the need for a constant network connection. Within this architecture, our Network Intrusion Detection System (NIDS) model was converted to a TensorFlow Lite model to fit the computational limitations of edge devices and functions. The NIDS monitors the incoming data for anomalies or indicators of potential cyber threats, such as atypical machine operations or unauthorized access attempts. When a possible threat is identified, the edge device triggers measures to mitigate the risk. After the initial stages of threat detection and data processing, the data is sent to the fog layer. This layer adds more processing power and storage, helping to refine the data and make sense of it. After this step, the refined data are moved to the cloud for more complex analysis, long-term storage, and historical trend analysis. The cloud provides the necessary resources for thorough analysis and decision-making.

By integrating XAI with the help of SHAP into this system, the decision-making process of the NIDS model can become more transparent and easier to understand. Applying SHAP in the edge computing structure can provide clear and helpful explanations for every detected intrusion. If the edge device detects a possible threat, the information is sent to the fog layer for XAI analysis. The fog layer is used for the XAI analysis because of the computational limitation in the edge devices. The explanations can help technicians and analysts respond quickly and accurately to possible threats. They also help to improve the IDS over time. Feedback from SHAP's insights can improve accuracy, reliability, and security against

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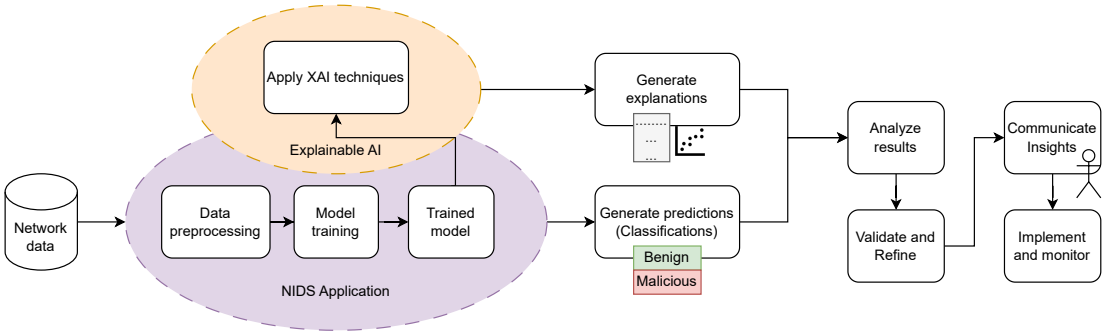


Fig. 2: Our intrusion detection workflow with explainable AI

cyber threats in the IIoT environment.

IV. Method

This study incorporates SHAP with deep learning models in IIoT environments, enhancing transparency while maintaining real-time detection efficiency in resource-constrained edge devices. Fig. 2 illustrates the workflow used. The process begins with data preprocessing, including cleaning and splitting network data for model training. The deep learning models (CNN, LSTM, and GCN) are then trained to classify the network traffic as benign or malicious, addressing different aspects of intrusion detection. These models generate real-time predictions that enable the timely identification of threats essential to maintain network security. SHAP was then applied to the predictions to make the model decision-making processes transparent. It assigns importance values to the features for each prediction, explaining the reasoning behind the decisions. A global SHAP analysis, visualized through summary plots, identified key features influencing overall model behavior. Local SHAP analyzes, using force plots and detailed contribution of features to specific predictions offered more actionable insights. A heatmap compared the importance of features in the models, providing a comparative perspective. Finally, the impact of integrating SHAP was evaluated by measuring changes in accuracy, computation time, and inference time per instance to confirm that the addition of XAI maintained the effectiveness of the models in intrusion detection while achieving a balance between transparency and performance.

V. Implementation

This section outlines the hardware setup, dataset, model architectures, integration of XAI techniques, and model conversion steps used to implement IDS in an edge environment.

A. Hardware setup

The implementation relies on a central computer and a Raspberry Pi (RPi) to simulate edge devices. The central computer handles initial model training using a high performance environment. The trained models are converted to the TFLite Flatbuffer format, optimized for resource-constrained devices. These models are deployed and evaluated on the RPi for accuracy and execution time under conditions that replicate industrial environments.

TABLE I: Description of key features

Feature Name	Description
`ct_srv_dst`	Count of services for destination
`ct_state_ttl`	Count of state and time-to-live values
`sttl`	Source to destination time to live
`dttl`	Destination to source time to live
`swin`	Source TCP window size
`stcpb`	Source TCP base sequence number
`ct_dst_ltm`	Number of connections of the same source address
`proto`	Transaction protocol
`dpkts`	Destination packets
`sload`	Source bits per second
`synack`	TCP connection setup time

B. Dataset

The UNSW-NB15 dataset, developed by the Cyber Range Lab at the Australian Centre for Cyber Security, serves as a benchmark for evaluating intrusion detection systems. By incorporating realistic and diverse network traffic scenarios, it addresses the limitations of older datasets [20], [21]. The dataset contains over 2.5 million records and 49 features, categorized into essential, content-based, time-based, and connection-based characteristics. Although all features are used in this study, a subset is listed in Table I. These selected features effectively capture critical aspects of network behavior, such as timing, protocol usage, and traffic volume. The dataset categorizes traffic into nine attack types (Fuzzers, Analysis, Backdoors, Denial of Service (DoS), Exploits, Generic, Reconnaissance, Shellcode, and Worms), as well as benign traffic, making it suitable for robust training and evaluation. To ensure compatibility and accurate interpretations, data cleaning was performed to remove irrelevant entries, followed by preprocessing to prepare the features for SHAP analysis.

C. Model architecture

All models are designed for binary classification, focusing on balancing accuracy and computational efficiency for deployment on edge devices. Each model targets specific data structures and provides its own approach to intrusion detection.

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TABLE II: CNN configuration

Type	Units	Activation function	Parameters
Conv1D	64	ReLU	256
MaxPooling1D	-	-	0
Flatten	-	-	0
Dense	50	ReLU	57 650
Dense	1	Sigmoid	51
Hyperparameter	**Value**		
Optimizer	Adam		
Batch size	256		
Epochs	50		

TABLE III: LSTM configuration

Type	Units	Activation function	Parameters
LSTM	64	Tanh	16 896
Dense	1	Sigmoid	65
Hyperparameter	**Details**		
Optimizer	Adam		
Batch size	256		
Epochs	50		

The architecture of each model is optimized for lightweight implementation, ensuring effective real-time detection while maintaining computational efficiency.

1) CNN: The architecture of the CNN model, shown in Table II, captures spatial patterns in the network traffic data. A 1D convolutional layer with 64 filters extracts spatial features, followed by max-pooling to reduce dimensionality. The flattened layer feeds into two dense layers, with ReLU activation in hidden layers and sigmoid activation in the output layer for binary classification. This architecture effectively captures complex spatial hierarchies within grid-based data.

2) LSTM: The LSTM architecture, detailed in Table III, is designed for sequential data, capturing temporal dependencies in network traffic. It uses tanh-activated memory cells to retain and process information in multiple time steps, making it ideal for detecting patterns that unfold over time, such as slow-developing attacks in IIoT systems.

3) GCN: The GCN model, as shown in Table IV, processes graph-structured data, making it especially effective for detecting coordinated attacks involving multiple network entities. The architecture includes graph convolutional layers to collect information from neighboring nodes, followed by dense layers for the final classification. ReLU activation ensures non-linearity, and dropout layers mitigate overfitting, enabling the model to analyze relational dependencies between devices in IIoT environments.

D. Integration of XAI Techniques

SHAP values were incorporated into the CNN, LSTM, and GCN models. For CNN, SHAP's 'DeepExplainer' was implemented, as it is optimized for neural networks with layer-based structures, making it effective for analyzing spatial feature contributions. The 'KernelExplainer' was applied to the LSTM model to interpret temporal dependencies in sequential data. Lastly, the 'Explainer' was chosen for the GCN due to its suitability for graph-structured data and relational feature analysis. This customized selection ensured meaningful and accurate explanations aligned with each model's architecture.

TABLE IV: GCN Configuration

Type	Units	Activation Function	Parameters
GCNConv	16	ReLU	768
Dropout	-	-	0
GCNConv	2	-	34
Hyperparameter	**Value**		
Optimizer	Adam		
Epochs	50		

The integration provided a transparent view of how each feature influenced the model predictions. The impact of integrating SHAP on the model's performance was evaluated by comparing the pre-and post-XAI accuracy and execution times.

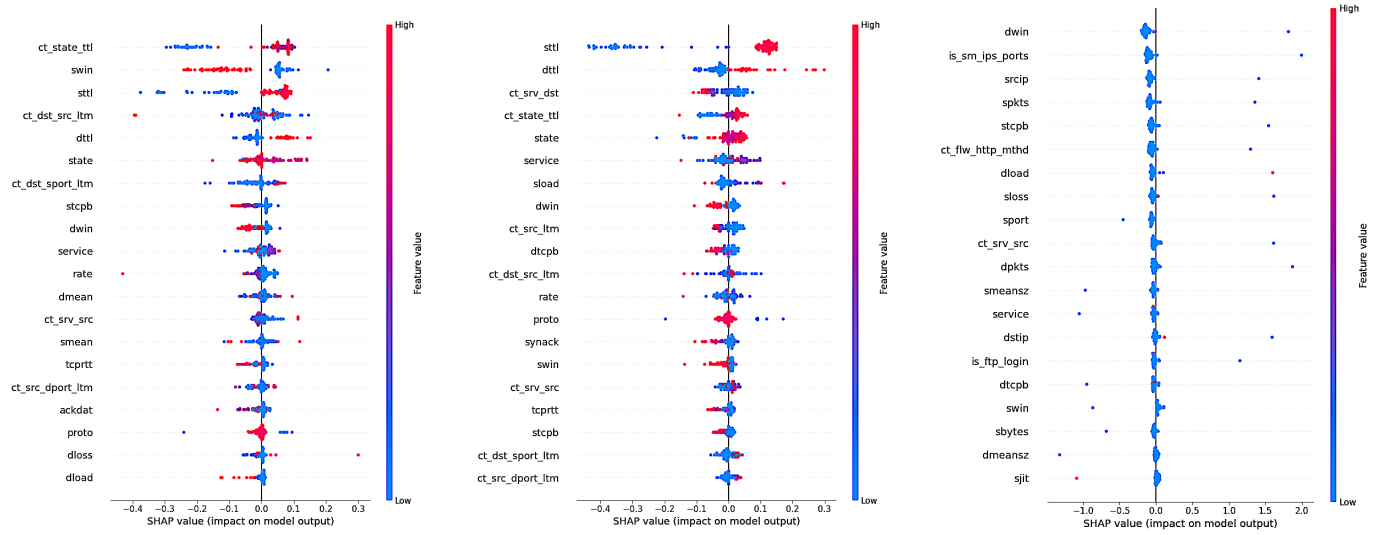
E. Model Conversion to TensorFlow Lite

All models undergo a conversion to TensorFlow Lite for deployment on the RPi. The CNN and LSTM models, originally created in Keras, were directly transformed using TensorFlow Lite's TFLiteConverter, which allows for int16 activations and int8 weights to improve optimization. For the GCN model, developed using PyTorch, it required an intermediate conversion step to Open Neural Network Exchange (ONNX) before being processed by TensorFlow Lite, to ensure compatibility and efficient performance on the RPi.

VI. Result

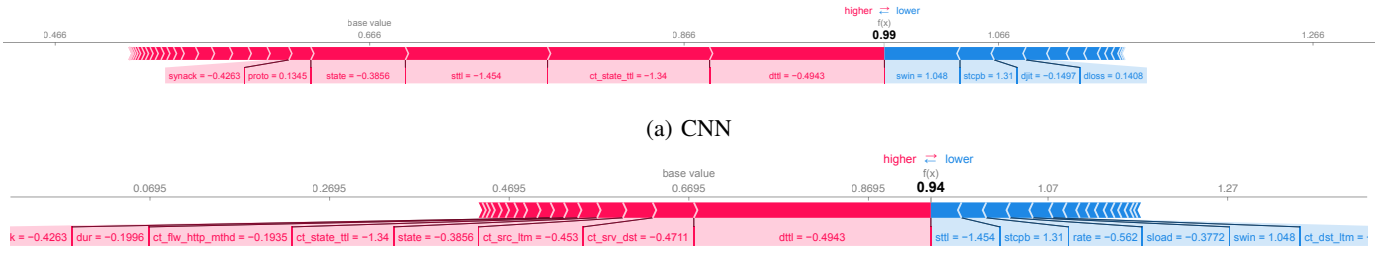
This section presents the findings of applying XAI techniques to deep learning models for intrusion detection in IIoT environments. The results are structured to demonstrate the effectiveness of integrating SHAP values with CNN, LSTM, and GCN models, highlighting their interpretability and performance. In Fig. 3, the SHAP summary plot highlights the key characteristics that contribute to the prediction of the model, illustrating the distinct strengths of CNN, LSTM, and GCN in interpreting network traffic. It highlights the importance and impact distribution of each feature, making it helpful in understanding the model's general behavior and the features significance. The features are ranked on the y-axis, with the most influential at the top and the least influential at the bottom. Their importance is quantified by the average magnitude of their SHAP values across all predictions, indicating their relative impact on the model's decision-making process. Each point in the plot represents a SHAP value for a feature for a single prediction. The points on the x-axis indicate the influence of a feature. The greater distance from the center line (0) reflects a stronger impact. The x-axis represents the SHAP values, which quantify the impact of a feature on the model's output. A positive SHAP value indicates that the feature pushes the prediction towards a higher value (in this case, indicating attack behavior). A negative SHAP value pushes it towards a lower value (indicating benign behavior). The color of each point reflects the feature value, with blue representing low values and red representing high values. The color coding illustrates how feature values affect the predictions. High feature values (red) associated with positive SHAP values indicate that the feature contributes significantly to predicting

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(a) CNN (b) LSTM (c) GCN

Fig. 3: SHAP summary plot for the models



(a) CNN

(b) LSTM

Fig. 4: The impact of individual features on the model's prediction for a specific observation.

an attack. Low feature values (blue) with negative SHAP values contribute to predicting benign behavior.

A. CNN

Fig. 3a shows a summary plot of the CNN model and its most essential variables for establishing a classification. The CNN model demonstrates a strong emphasis on spatial features such as `ct_state_ttl` and `sttl`, which are critical for detecting anomalies in localized network patterns. These findings suggest that CNNs are particularly effective in identifying static network behaviors and can be further optimized by improving feature selection pipelines tailored for spatial analysis.

Fig. 4a shows how individual features influence the model's prediction for a specific observation. It is helpful for a detailed case-by-case analysis. In this plot, the base value represents the model's baseline prediction without considering any features. The features are listed horizontally, each accompanied by a bar that shows its SHAP value, indicating its contribution to the final prediction. Features with positive SHAP values are shown on the right side of the baseline. These features increase the prediction value, pushing the prediction towards a higher value, which indicates a likelihood of attack behavior in this context. Features with negative SHAP values are shown on the left side of the baseline. These features decrease the prediction value, pushing the prediction towards a lower value, indicating benign behavior. The contribution of each feature is visualized as a bar whose length represents the magnitude of the impact. The color of each bar indicates whether the feature's value is high (red) or low (blue), providing an additional layer of understanding regarding the influence of feature values. For example, a high feature value that increases the prediction (red bar on the right) suggests that the feature strongly indicates an attack. In contrast, a low feature value that decreases the prediction (blue bar on the left) suggests that it is a sign of benign behavior. The analysis shows that `swin` and `stcpb` have significant positive contributions, allowing the model to classify the traffic as malicious. In contrast, features such as `state` and `dttl` exhibit negative SHAP values, pushing the prediction toward a benign classification.

B. LSTM

The SHAP summary plot in Fig. 3b provides an overview of the impact of each feature on the output of the LSTM

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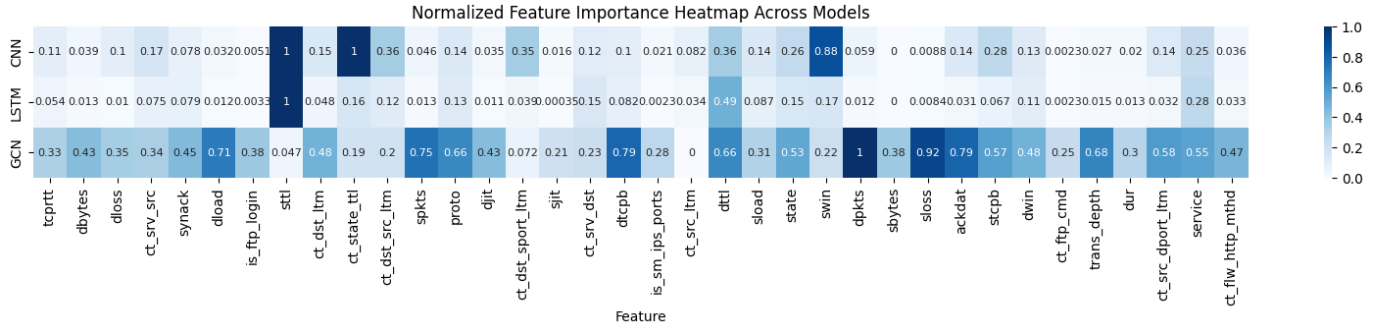


Fig. 5: Heatmap of normalized feature importance across the three models

model. The feature `sttl` has the highest impact on the model output, with high and low values significantly affecting the prediction. `dttl`, also has a notable impact, indicating the importance of TTL features in network traffic analysis. The feature `ct_srv_dst` shows a varied impact, with higher values generally leading to a higher prediction probability. `ct_state_ttl` is another influential TTL feature, underscoring its importance in different temporal aspects of the data. These results confirm the model's suitability for detecting sequential anomalies, such as slow-evolving attacks. This underscores the importance of using LSTMs in environments where time-sensitive threats are common.

The SHAP force plot in Fig. 4b visually explains the individual predictions. The contribution of each feature increases or decreases the model output. The plot highlights that `stcpb` and `swin` have notable positive contributions, pushing the model's output towards a higher probability of intrusion. However, features such as `ct_src_ltm`, `sttl`, and `dttl` have negative contributions, decreasing the likelihood of an intrusion being detected. This detailed view allows us to understand how each feature influences specific predictions, enhancing the models' interpretability. This demonstrates the ability of the LSTM model to weigh temporal dependencies and adjust predictions based on the sequential nature of network traffic data.

C. GCN

For the GCN model, the SHAP summary plot in Fig. 3c, shows a different pattern of feature importance compared to the CNN and LSTM models. The GCN model, designed to capture relational dependencies within graph-structured data, emphasizes the `dpkts` and `sload` features. These features are essential for understanding the relational context within network traffic, making the GCN model especially effective in scenarios where the relationships between different data points play a critical role in detecting coordinated attacks. Unlike the CNN and LSTM models, the GCN's SHAP values are more dispersed, reflecting the model's broader consideration of the network traffic's relational structure. This suggests that the GCN model captures complex interactions between features, which may be less apparent in traditional flat-data models.

D. Comparative Feature Importance Analysis

The heatmap in Fig. 5 illustrates the importance of normalized characteristics in the CNN, LSTM, and GCN models. The heatmap shows the contribution of each feature to the model's predictions, normalized for comparison purposes. The CNN model places significant importance on `ct_state_ttl`, `sttl`, and `'ct_dst_ltm'`. These features are essential for recognizing complex spatial patterns within network traffic data. The features `proto` and `dpkts` also show high importance, reflecting the model's ability to capture various aspects of network data. The LSTM model also highlights `sttl` as the most critical feature. This model emphasizes temporal dependencies, as shown by the values of `ct_state_ttl`, `dttl`, and `swin`. These features are crucial for capturing sequential patterns over time and for detecting temporal anomalies in network traffic. The GCN model places the highest importance on `dpkts`, followed by `sload` and `synack`. The feature `sttl` is much less important than the other models. This shows that the GCN model concentrates on understanding relational dependencies within the data, using graph structures to identify interactions between data points. The heatmap provides insight into the importance of the features in different models. Although some features such as `dttl` are essential for all models, others show varying degrees of influence, depending on the model architecture.

E. Performance Impact of XAI Integration

To evaluate the suitability of the models for deployment in resource-constrained environments, we performed performance measurements using a RPi 4 Model B, representing an edge device. Each model (CNN, LSTM, and GCN) was evaluated using approximately 80,000 samples from the UNSW-NB15 dataset, both with SHAP-integrated and without using SHAP. For each configuration, we performed 100 independent inference runs to compute the average accuracy and mean inference time per instance. In Table V, we present a comparative analysis of the performance metrics of the models, highlighting the trade-offs between accuracy, interpretability, and computational cost. The CNN model demonstrated the fastest inference time, which makes it well suited for edge device deployment. The GCN model achieved the highest accuracy with slightly longer inference times. The LSTM model

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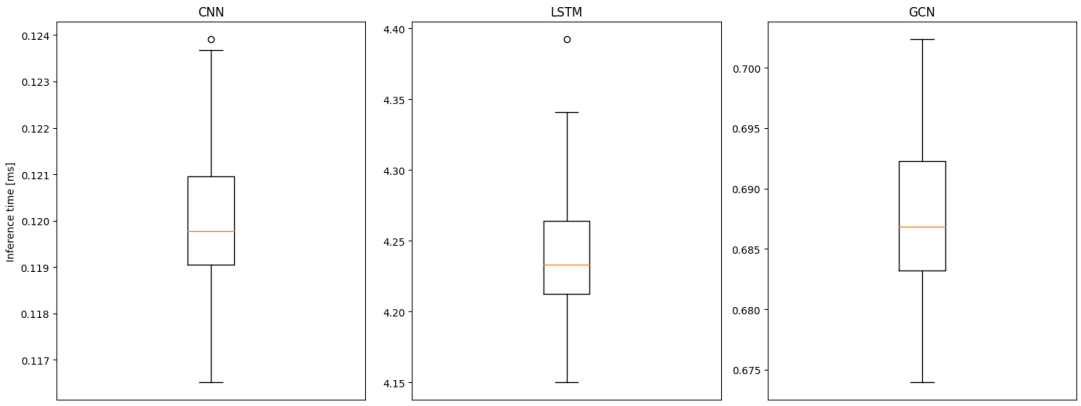


Fig. 6: Comparison of inference times for the models after SHAP integration

TABLE V: Accuracy and average inference time per class before and after SHAP integration

Model	Accuracy	Avg/Class [ms]	Avg/Class w/ SHAP [ms]
CNN	0.937	0.115	0.120
LSTM	0.932	3.902	4.236
GCN	0.975	0.667	0.688

showed the longest inference times, which is expected given its sequential data processing in a constrained environment, but maintained comparable accuracy.

Fig. 6 compares the distribution of inference times for each model. The box plot highlights the differences in the variability of inference time, where CNN shows the most consistent and fastest performance. LSTM shows a broader range of inference times, reflecting its higher computational cost. The GCN model demonstrates inference times between CNN and LSTM. The integration of SHAP added some computational overhead across all models, as reflected in the mean inference time per class. The CNN model showed the smallest increase in computation time, indicating its robustness in maintaining efficiency with interpretability. The LSTM model, on the other hand, experienced the most significant increase in inference time with SHAP, highlighting the additional cost of explaining sequential dependencies. The inference time of the GCN model also increased, though to a lesser extent than that of the LSTM, because of the complexity of analyzing relationships within graph-structured data.

VII. Discussion

The analysis of CNN, LSTM, and GCN models highlights their individual strengths and contributions to intrusion detection in CPS and IIoT environments. Across all models, the features `ct_state_ttl` and `sttl` emerged as critical indicators of anomalies, demonstrating their importance for detecting network irregularities. However, each model's architecture emphasizes different aspects of the data. CNNs excel at identifying spatial patterns, focusing on features like `ct_dst_ltm` and `swin`, which reflect localized behavior. In contrast, LSTMs capture temporal dependencies, making them effective in analyzing features such as `ct_srv_dst` and `dttl`, which are associated with event sequences. GCNs prioritize relational dependencies, highlighting features like `dpkts` and `sload` that describe interactions between network nodes. The shared significance of `ct_state_ttl` and `sttl` across all models suggests that a unified feature set could enhance IDS frameworks by supporting multiple models optimized for diverse deployment environments. For example, CNNs could focus on static networks, while LSTMs and GCNs could address time-sensitive or multinode scenarios. A hybrid approach combining these strengths offers the potential to improve detection accuracy and adaptability. These hybrid architectures might incorporate CNNs for the initial extraction of spatial features, which are then input into an LSTM to understand temporal relationships, while GCNs are utilized to examine relational patterns. This multilayered setup could enable IDS systems to identify a broader spectrum of attack vectors with improved accuracy.

As a response to research question 1, regarding how explainable AI techniques can be integrated into lightweight deep learning models for IDS in CPS and IIoT environments, this study demonstrates that SHAP can be effectively incorporated with lightweight CNN, LSTM, and GCN models deployed on edge devices. SHAP values provide feature-level explanations for both global and local model behavior, offering transparency without sacrificing efficiency. The integration process involves training the models on network traffic data, applying SHAP to analyze their outputs, and generating interpretable explanations that help system administrators make informed decisions.

To address research question 2, concerning the impact of incorporating explainable AI on the accuracy, trustworthiness, and efficiency of IDS, the study finds that SHAP significantly improves transparency and trustworthiness. System administrators can better understand and trust predictions, and integration does not compromise the accuracy of the model. Although SHAP introduces minimal computational overhead, inference times vary between models, with CNNs being the fastest, GCNs slightly slower, and LSTMs the most computationally demanding. This variability underscores the

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need to align model selection with deployment requirements, particularly in time-critical environments.

Future work could explore additional XAI techniques, such as LIME or Grad-CAM, to complement SHAP and provide alternative perspectives on model interpretability. Hybrid architectures combining CNNs, LSTMs, and GCNs could be further investigated to leverage their strengths in spatial, temporal, and relational pattern recognition. While this study demonstrates the feasibility of integrating XAI into lightweight IDS, the results are based on the UNSW-NB15 dataset and a specific edge hardware platform, which may affect generalizability. Testing these models on larger and more diverse IIoT networks, as well as different hardware configurations, would help evaluate scalability and robustness under real-world conditions. Testing these models on larger and more diverse IIoT networks would help evaluate scalability and robustness under real-world conditions. Federated learning offers another promising avenue, enabling decentralized training across multiple edge devices to preserve data privacy while enhancing security. Efforts to reduce false positives through refined feature selection and advanced anomaly detection techniques could further improve IDS reliability. Furthermore, exploring reinforcement learning or adaptive models that evolve with emerging threats could ensure that systems remain effective in dynamic environments. Finally, addressing ethical considerations, such as balancing explainability with user privacy and security, will be vital to building trust in these solutions across various CPS and IIoT sectors.

VIII. Conclusion

This study integrates XAI techniques into lightweight deep learning models for intrusion detection on IIoT edge devices, demonstrating that SHAP can provide information on model behavior. In doing so, we have addressed the black-box nature of these models and, by that, enhanced their transparency and trustworthiness. This research has shown that integrating XAI techniques with lightweight deep learning models improves the overall efficacy of IDS in CPS and IIoT environments by improving transparency without sacrificing accuracy or efficiency. This work contributes to creating secure and trustworthy IIoT ecosystems, ensuring compliance with regulatory frameworks like the EU AI Act, while improving the overall effectiveness of IDS. By integrating XAI into IIoT-based IDS, this work not only improves the model explainability but also improves the ability to quickly identify and mitigate security threats in industrial environments, strengthening cybersecurity in real-time systems.

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